



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS8501 Theory of Computation

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit III / Pushdown automata

Course Outcome: CO 3

Topic Learning Outcome: TLO 9

Activity Chosen: Reflection

Justification:

- The reflection activity takes place immediately following the discussion of a specific topic, letting learners to analyse their own understanding of the concept. Knowing the fundamentals of Pushdown Automata is somewhat confusing, such as when to push and pop the symbols into the stack, and students cannot solve PDA to CFG and vice versa without a strong understanding. The instructor can assess the students' level of understanding with the help of this activity. This also provides an opportunity to determine the level of student understanding in the specific topic, and the activity helps in resolving of doubts.
- **Time Allotted for the Activity:** 15 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes.
- In order to address the doubts and to know the level of understanding of the students, they were insisted to honestly describe the concepts that they didn't understand well.
- The students were asked to write what they have understood from the topic and also insisted to write if they have any queries about the topic.
- The instructor uses the written material of each student to assess their understanding on the topic.
- The instructor responds to the students' comments the next day in class.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
CO 1	2	2	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
	2	2	2	1	1	1	1
Justification for correlation	Apply knowledge of context free grammar in the construction of PDA	To analyze complex engineering problems using PDA	To design PDA	To design the appropriate PDA for given CFG	To work as an individual	To Communicate effectively on complex engineering activities	To design the PDA, students will be able to develop various software components

• **Images / Screenshot of the practice:**

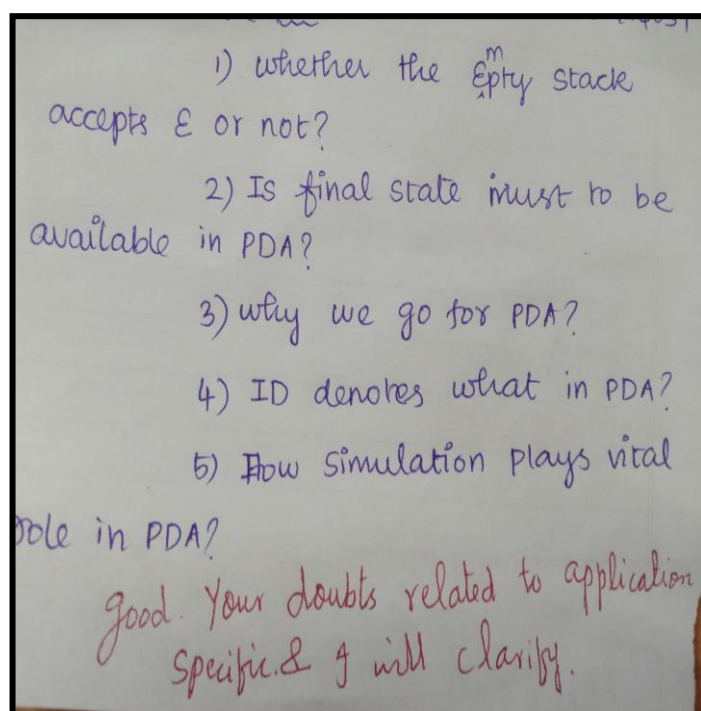
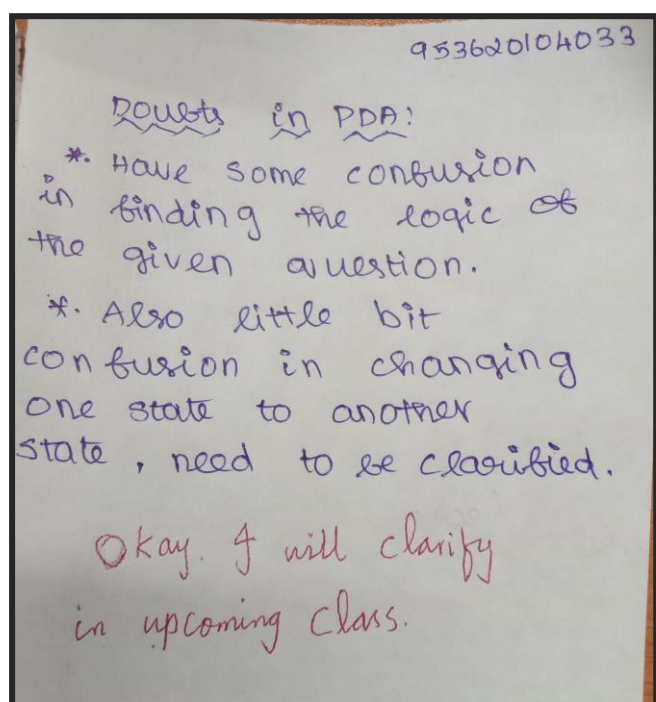


Fig 1: Sample reflection activity

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students stated that the activity assisted them in determining their level of understanding of the concept.
 - Students informed the instructor that the activity motivates them to pay attention in class and ask questions.

- ***Benefit of the practice:***

- The activity helped the instructor to analyze, evaluate and improve their own learning.
 - Reflection activity helped the instructor to identify the students level of understanding and also helps to plan the next session accordingly.

- ❖ ***Challenges faced in implementation:***

- With the exception of a few students, the majority of the students actively participated. They are not involved in sharing their level of understanding or raising questions.
 - Motivate students who are not actively participating in the activity by highlighting the benefits of reflection and creative thinking.

References:

1. <https://www.clemson.edu/otei/documents/Reflection%20Activities%20r.pdf>
2. <https://www.ritrjpm.ac.in/images/computer-science/Reflection-HCI-MSS.pdf>
3. <https://www.ritrjpm.ac.in/images/computer-science/SM-UII-AI-CS8691.pdf>
4. <https://teaching.utk.edu/wp-content/uploads/sites/78/2018/04/ReflectionActivities.pdf>

Signature of Faculty Member

HOD